



Horizontalism: a critique

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This article offers a critique of the horizontalist view of money that banks are passive in the face of credit demand. It is argued that banks' liquidity preference influences their responsiveness to the demand for credit. Their liquidity preference is expressed in risk assessment (understood in terms of Keynes' theory of uncertainty). It is argued that rationing in the sense of adverse changes in risk assessment occurs systematically in the downturn of the business cycle. Systematic rationing also occurs with respect to particular classes of borrowers; the focus here is on the case of small firms.

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Introduction

Post-Keynesian monetary theory has become identified with the notion that the money supply is endogenous, and credit-driven (see for example Arestis, 1988). This notion derives from a focus on the process by which monetary aggregates are generated, and has in turn formed the focus of a critique of monetarist monetary policy.

But within this general theoretical perspective there are differences as to detail which are potentially significant both for theory and for policy (see Lavoie, 1984; Wray, 1990, ch. 5). The endogenous money standard has been borne most notably by Kaldor (1986) and Moore (1988); their position is characterised by Moore as 'horizontalist', referring to a perfect elasticity of supply of credit. As Moore (1988, p. xii) puts it: 'modern commercialised banks are price setters and quantity takers in both their retail deposit and loan markets'. This implied passivity of banks in the credit market is a source of some of the differences within post-Keynesian monetary theory. Further difference of opinion stems from the rejection by both Kaldor and Moore of Keynes's principle of liquidity preference. Indeed, some have pointed to the importance of considering the liquidity preference of banks, which might encourage credit rationing (see Dow, 1982; Dow and Earl, 1982; Wray, 1990; Heise, 1992; Dow, 1993). It can be argued that this extension of liquidity preference theory to bank behaviour is not compatible with Kaldor's and Moore's overall stance.

It is the purpose of this article to offer a partial examination of horizontalism from an endogenous credit/liquidity preference standpoint. The focus here will be on the liquidity preference only of the banks. A complete account would consider the liquidity preference of all sectors (as offered in Dow and Dow, 1989). Further, the issue is not addressed directly of whether endogenous credit theory is, as Kaldor and Moore suggest, incompatible with liquidity preference theory in its conventional application to individual

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investors. Wray (1990, 1992) provides a full examination of the relationship between endogenous credit, liquidity preference and the demand for money in relation to the rate of interest.

In the next section, the theoretical case is made for amending the horizontalist position to allow for systemic credit rationing, referring particularly to the business cycle. It is argued that there may also be systematic rationing of particular classes of borrowers. Small firm finance is considered in the third section as one such counter-example to the banks-as-quantity-takers hypothesis. It is impossible with current data availability to quantify adequately the extent of rationing. The purpose here is the more limited one of demonstrating that systematic rationing occurs; the pro-horizontalist evidence, that overdraft facilities are not fully used up, is questioned on the grounds that the availability of overdraft facilities itself may be rationed. This argument is offered fully in the spirit of the endogenous credit approach; it is offered as an attempt to qualify and enrich the horizontalist version of that approach.

Liquidity preference applied to bank lending behaviour

Kaldor and Moore (with varying degrees of explicitness) base their argument that the supply of credit accommodates demand on three factors. First, banking systems are adept at innovation, so that any attempts by the authorities to curtail credit can be subverted; this element is captured in Goodhart's Law, that an observed empirical relationship may break down as soon as the government attempts to exploit it for policy purposes. Even if the authorities succeed in controlling a particular monetary aggregate, innovation by banks can generate an increase in credit outside that particular aggregate, either elsewhere on the balance sheet or, increasingly, off the banks' balance sheet altogether (see Gardner, 1993). Second, modern banking systems function on the bedrock of the confidence generated by the lender-of-last resort function of the central bank. Banks are therefore free to supplement reserves to back new loans. Third, banks themselves have limited control over credit supply owing to the widespread practise of granting overdraft facilities or credit lines.

But, as Rousseas (1986, pp. 92–96) points out, the argument that banks innovate sufficiently smoothly and rapidly to allow them always to accommodate demand for credit has not been fully articulated. Wojnilower (1980) does demonstrate the increasing capacity of US banks to meet credit demand with banking deregulation. However, he also investigates periods of credit crunch, when the coming into force of some regulatory limit induced banks to contract credit supply. The implication is that the authorities can discourage credit demand accommodation if they wish to do so; but it requires a substantial jolt to the banking system to which innovation cannot adequately respond (in the short term at least).

In the absence of such intervention, would banks normally choose to accommodate all demand for credit anyway? Is recourse to the lender-of-last-resort facility only limited by the demand for credit? This question is also fundamental to the argument that borrowers determine the supply of credit through overdraft facilities or credit lines; are these facilities supplied on demand?

Neither Kaldor nor Moore would seem to argue that literally all demand for credit is accommodated. Kaldor (1986, p. 13) quotes as follows from the Radcliffe Report, without apparent demur:

The ease with which money can be raised depends on the one hand upon the composition of the spender's assets and on his borrowing power and on the other hand upon the methods, moods and resources of financial institutions, and other firms which are prepared (on terms) to finance other people's spending. (Radcliffe, 1959, para. 389)

The factors which may limit acceptance of loan requests are noted by Moore in a manner more circumspect than Radcliffe, but not as we shall see incompatible with him:

Commercial bank loan officers must ensure that loan requests meet the bank's income and collateral requirements. They must in general satisfy themselves as to the credit-worthiness of the project and the character of the borrower. (Moore, 1988, p. 24)

The door has thus been opened for considering the possibility of credit-rationing, although neither Kaldor nor Moore follows up the significance of the possibility. The 'moods' of financial institutions may become more pessimistic, or display reduced confidence in prediction. Then borrowers who had previously been acceptable may find the value of their collateral and projections of future income streams reduced according to the assessment by financial institutions. The borrowers themselves may perceive no change in their own assessment of their credit-worthiness. It is a matter of semantics whether or not the resulting availability of credit is termed 'rationing'. The important point is that it is no longer the case that all demand for credit is met. Since the 'moods' of financial institutions are likely to follow a pro-cyclical pattern, so too will availability of credit.

Insofar as increased perceived risk is incorporated in a risk premium, borrowers may be rationed by price. Indeed, the evidence suggests that interest rate spreads vary counter-cyclically (Rousseas, 1986, ch. 3; Davis, 1992, ch. 6) while the debt/equity ratio varies procyclically (Davis, 1992, ch. 2). Indeed, Davis demonstrates that financial crises occur following periods of rapid credit expansion when banks unduly reduce risk premia, buoyed up by animal spirits; with the onset of crisis comes a sharp rise in spreads as risk premia rise, and a sharp drop in borrowing. (See Davis and Henry, 1993, for a detailed analysis of spreads and their rationale.)

The 'resources' of financial institutions (as noted by Radcliffe) are also likely to follow a pro-cyclical pattern. It is capital adequacy ratios rather than reserve ratios now which are the operative constraint (see Gardner, 1993). Increasingly, capital adequacy is being applied as a regulatory requirement; higher ratios are required the higher the risk attached to the portfolio (see Hall, 1992). This brings us back to risk assessment; if perception of risk is counter-cyclical, then so is the required proportion of capital backing. However, the equity market is also subject to 'moods' which if anything are pro-cyclical. Therefore, the cost of capital is likely to rise and availability to fall in downturns, putting external constraints on bank credit expansion in downturns.

In fact, the timing both of banks' and of equity markets' perception of risk attached to banks' assets tends to lag significantly behind available evidence, so that the cyclical pattern is also lagged. For example, although the Federal Reserve and World Bank were aware of increased sovereign debt risk in the 1970s (see, for example, Sargan, 1976, 1977) it was only after Mexico's default in 1982 that banks appeared to absorb the available evidence of increased default risk. Then, there was a lag in the equity market's assessment of the risk attached to bank's portfolios. Only after Citibank took the lead by making provision for bad debts in 1987 did the market seem aware that the value of bank assets had been in jeopardy for years; only then did the market constrain the banks by reduced willingness to buy bank equity.

The horizontalist position already incorporates rationing by means of the interest rate determined by the monetary authorities. The qualification of the assertion that 'all demand is met' by the word 'credit-worthy' might be taken as simply adjusting the rationing interest rate by the appropriate risk premium. If risk assessment of borrowers by banks and equity markets were deterministic, then it could be argued that the horizontalist position would not be significantly affected; indeed the inattention of horizontalists to the concept of credit-worthiness can only be explained in this way. But the discussion above has suggested not only that risk-assessment is subject to uncertainty, and is thus prone to discrete shifts, but that shifts in risk assessment tend to follow systematic patterns which may have only a loose relationship with real economic developments.

As Keynes's (1973A) work on probability clearly implies, the future balance sheets of banks and their customers cannot be predicted using quantitative probability techniques. (The fact that banks may be forced into expressing default risk in quantified form does not prove anything about whether such a technique can be justified.) Under conditions of uncertainty (unquantifiable risk), the potential lender must fall back on habit, conventional judgement and assuming the future to replicate the past (see Keynes, 1973C, pp. 109–123). Both Lavoie (1984) and Heise (1992) have drawn attention to the crucial role of banks' expectations (both use the expression 'animal spirits') in determining the volume of credit.

If risk cannot adequately be determined using quantitative methods, the distinction between credit-worthy demand for credit and non-credit-worthy demand takes on heightened significance. In particular, since conventional judgement is more a macro than a micro phenomenon (see Hodgson, 1988, ch. 6), the issue of credit rationing and bank capitalisation ceases to be a purely micro phenomenon. Rationing as a result of conventional judgement as to credit-worthiness may then apply to whole classes of borrowers. Insofar as the banks themselves might fall into such a class under particular market conditions, and find general difficulty in raising capital to back increased lending, rationing of the banks themselves could become a general phenomenon.

The case study offered in the next section refers to a class of borrowers (small firms).¹ Here we will focus on the possibility of generalised credit rationing; we draw on the insightful work of Minsky (1975), which can readily be adapted to bank behaviour (see Dow and Earl, 1982, ch. 11). Minsky sets out a model of corporate financial behaviour, whereby the amount of external financing for investment projects depends on the availability of internal finance, the expected returns on the projects, borrower's risk and lender's risk. The last three are all judgments made under uncertainty. In particular, Minsky shows how the demand for and availability of external finance fall as borrowers' and lenders' perception of risk, respectively, rises. The demand for borrowed funds and the supply of borrowed funds are less interest elastic the greater is the perceived borrowers' risk and lenders' risk, respectively. The height of the credit demand curve reflects the proportion of expected returns which borrowers are willing to commit to debt servicing. Minsky shows how, other things being equal, perceived borrowers' risk rises as the level of borrowing rises, reducing the interest rate acceptable for any given level of borrowing. Similarly, the height of the supply schedule reflects the return required by lenders, including compensation for perceived risk. Minsky argues that perceived risk rises as volume of lending rises, other things being equal, but that increasing pessimism

¹ The experience of LDC debt over the last 2 years may provide an additional example of rationing of bank credit for a class of borrowers. (See Dow, 1993, ch. 11).

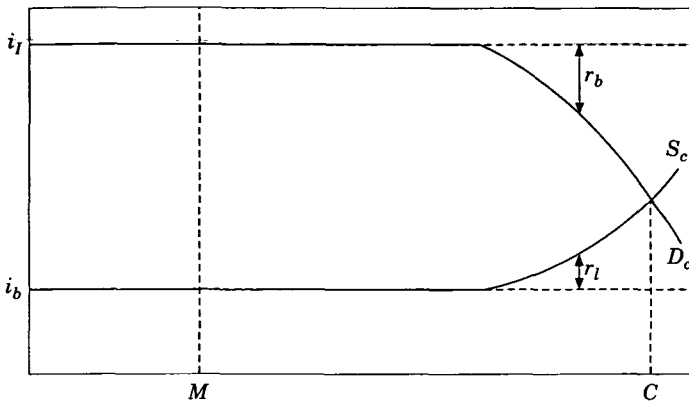


Fig. 1. Credit market with risk: expansionary phase.

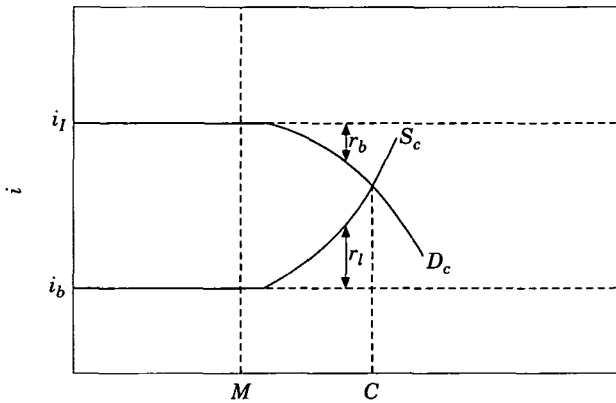


Fig. 2. Credit market with risk: contractionary phase.

about risk encourages banks to require a given risk premium at lower levels of lending. The supply curve becomes more inelastic the greater the perception of risk attached to any level of borrowing. The amount of credit and its price (including a risk premium) is thus determined by borrowers' and lenders' risk, given the availability of internal funds and the expected return on investment. Taken together, the four factors (with their propensity to vary systematically over the cycle) provide an account of credit expansion during an economic upturn as expectations become more optimistic, while the reverse is the case during downturns.

Adapting Minsky's framework along the lines of Dow and Earl (1982, pp. 140–141), we can construct a credit market diagram as shown in Figs 1 and 2 for two states of the economy. OM is the banking system's capital base; the greater the extent of lending relative to the capital base, the more fragile its portfolio. We shall assume for the moment that the capital base is adequate for all conceivable credit expansion. (We shall return to the question of the supply of capital to the banks below.) i_b is the base rate for bank lending, as determined by the central bank's discount rate. The supply of credit is determined by a mark-up on i_b , with the mark-up increasing owing to lender's risk, r_l as the volume of credit increases. The demand for credit is determined by the expected return on investment, i_f , and by borrower's risk r_b . Figure 1 represents the credit market

in an expansionary phase of the cycle, when perceived risk is low and expected returns high. The volume of credit as determined by the marginal supply of the credit curve is high. Figure 2 represents a recessionary situation, with low expected returns and high perceived risk, resulting in a lower volume of credit.

The i_b line may be seen as equivalent to the horizontalist supply of credit curve. However, bank loan charges are conventionally set within a rate structure (base rate plus risk premium).¹ The effective supply of credit curve is thus in fact Sc , which varies over the cycle with the degree of perceived risk.² A rate structure is implicit in 'the interest rate' shown in horizontalist credit supply curves. Here it is being suggested that there is systematic movement within the structure over the cycle which should be made explicit.

Risk assessment, as Moore points out, requires a valuation of collateral as well as a valuation of the planned project. During an economic downturn, valuation of collateral and of the expected returns on projects will be revised downwards. Once there is a collapse in asset values, credit crumbles (see Johnson, 1991, for US evidence of this phenomenon). But the revision of expectations occurs after a lag. Keynes (1972) first introduced the concept of credit crumble in 1931.³ In this article, Keynes refers to the conventional nature of bank expectation formation:

A 'sound' banker, alas! is not one who foresees danger and avoids it, but one who, when he is ruined, is ruined in a conventional and orthodox way along with his fellows, so that no one can really blame him. (Keynes, 1972, p. 156)

Once it becomes apparent to the banks that the value of their existing assets is falling, not only will this be reflected in the valuation of collateral offered by new potential borrowers, but, Keynes argues, the banks will seek to increase liquidity by curtailing new lending:

the banks, being aware that many of their advances are in fact 'frozen' and involve a larger latent risk than they would voluntarily carry, become particularly anxious that the remainder of their assets should be as liquid and as free of risk as it is possible to make them. This reacts in all sorts of silent and unobserved ways on new enterprise. For it means that the banks are less willing than they would normally be to finance any project which may involve a lock-up of their resources. (Keynes, 1972, pp. 153–154)

This passage contains a statement of liquidity preference on the part of banks. Written when reserve requirements might have been regarded as more of a constraint, the mention of 'resources' could be taken as reflecting the conventional (yet controversial) view of Keynes, that he took the money supply as fixed. Then the statement is equivalent to one of liquidity preference with respect to a given portfolio, as Keynes himself originally presented it (Keynes, 1973B, p. 166).

But putting the statement in an explicitly endogenous-credit context gives it added force. Providers of funds to banks (through the interbank or equity markets) will likewise perceive a collapse in the value of the banks' assets and will endeavour to maintain their

¹ There is evidence also of banks differentiating markets by means of differentials in fees for services (see McKillop and Hutchinson, 1990). With the increasing focus of bank activity on provision of services relative to provision of credit, fee structures will require increasing attention.

² The actual base rate would also be influenced by liquidity preference. If liquidity preference were high, there would be upward pressure on market interest rates. But for comparability with the horizontalist argument, we assume that the central bank acts in such a way as to hold the base rate steady.

³ Johnson (1991, p. 40) incorrectly cites Fisher as the originator of the concept of credit crumble, in his work on debt deflation; see Fisher, 1933.

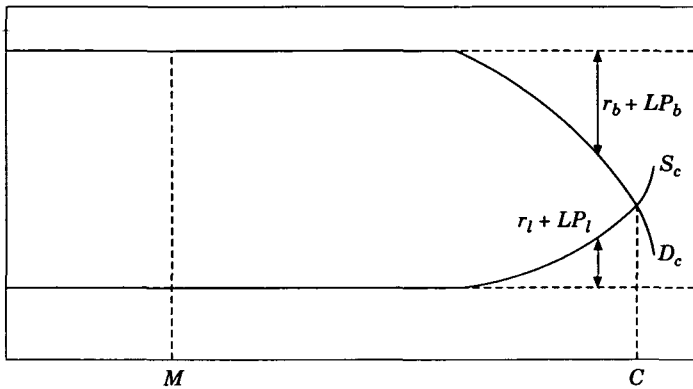


Fig. 3. Credit market with risk and liquidity preference: expansionary phase.

own liquidity by avoiding further commitment to the banks. Once banks find it difficult to fund loans and/or fall short of their capital-adequacy requirements, the overall size of their portfolios is limited. As a result there will be additional pressure to limit new lending. In confirmation of this viewpoint, Johnson (1991) has found the collapse in asset prices and the shortage of bank capital to have contributed significantly to the contraction in corporate lending in the US during 1989–1990. Supply factors are explicitly isolated from the effect of changes in economic activity which could be expected to affect credit demand (as is the case in other such studies, see for example Fazzari *et al.*, 1988).

This argument suggests that the supply of credit may well be close to horizontal through much of the business cycle, except for some classes of borrowers who routinely fall foul of the risk-assessment process (see the following section). But during a downturn the value of collateral and the prospective returns on investment projects fall. After a recognition lag, banks will respond by curtailing new lending in order to increase the liquidity of their balance sheets. There will be a further recognition lag on the part of providers of wholesale funds and capital to the banks, as they gradually perceive the value of the banks' own collateral falling; the ensuing cut-back in provision of funding and new capital to the banks will reinforce the contraction of credit.

The upward slope in the credit supply function in Figs 1 and 2 may then be seen, not as a measure solely of risk, but also of liquidity preference (LP). In extreme cases the curve may become vertical, when banks put an absolute ceiling on credit supply, placing free reserves instead in investments as better providers of liquidity.¹ The diagrams are shown in Figs 3 and 4 explicitly incorporating liquidity preference in this way. In addition, account is taken of capital adequacy requirements, and the possibility that capital availability relative to desired credit expansion is constrained in a recession. This is shown by OM shifting in along with the credit market curves in the recessionary situation in Fig. 4.

But there is a further layer to the process which complicates the argument. Credit is created not only to finance productive investment but also to finance speculation. Thus, a collapse in returns to speculation could itself initiate an increase in default risk and a reduced capacity (or willingness) to acquire equity in banks. Credit availability may then be curtailed because of increased liquidity preference on the part of the banks,

¹ A credit ceiling of this sort was envisaged for local (as opposed to national) financial markets in terms of a cut-off in the horizontal supply of credit curve in Dow (1987).

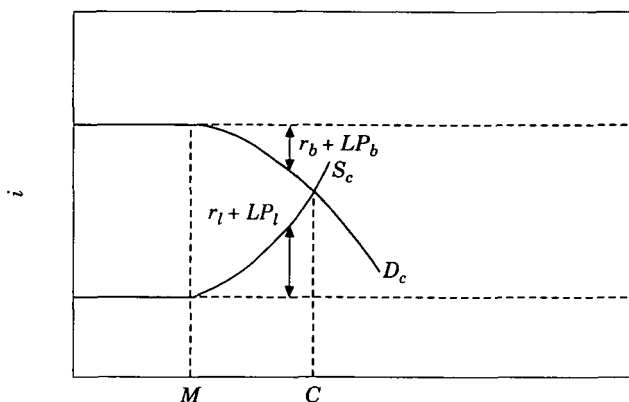


Fig. 4. Credit market with risk and liquidity preference: contractionary phase.

irrespective of the 'true' risk attached to commercial lending. Credit rationing for business would then arise quite independently of the expected returns on capital investment projects (Chick, 1992, ch. 11 and 12; Dow, 1993, ch. 4). In Marxian terms, this argument suggests, contrary to the horizontalist position, that financial crisis can be the forerunner of production crisis.

The horizontalist position, particularly as expressed by Moore, rests heavily on evidence of unused overdraft facilities (see for example Moore, 1988, pp. 24–26). Moore cites the relatively high proportion of loans in the US made under credit lines, and the extent of credit available but unused under credit lines. Reference is also made to the long tradition of overdraft facilities in the UK banking system. However, this tradition appears to be breaking down. The proportion of total loans made under overdraft facilities in Britain was 22% in 1984, when the breakdown was first published in the *Bank of England Quarterly Bulletin*; this proportion had dropped to 14% by 1992. There may well be, as in the US, significant unused facilities. But the question has not been addressed as to whether there is rationing of overdraft facilities themselves. When we come in the next section to consider the case of small firms, it becomes apparent from the US evidence that there is indeed significant rationing of this form. It is impossible to quantify rationing of this sort. But if there is indeed systematic rationing, even of sub-classes of borrowers, then it can no longer be maintained, without explicit qualification, that banks are quantity takers.

Small firm finance: is there a credit gap?

In this section we examine the case of small firms as a possible example of credit-rationing with respect to particular classes of borrowers. There is a growing literature on 'equilibrium' credit rationing, i.e. rationing which is rational for profit-maximising banks (see Driscoll, 1991 for a review). This rationing is explained in terms of moral hazard if high (market-clearing) interest rates were to be charged on all loans, and adverse selection whereby high (market-clearing) interest rates would deter risk-averse borrowers; in both cases the average degree of risk-aversion among borrowers would fall unacceptably to the bank. These problems could arise because of the convenience of not having a rate structure fine-tuned to each individual borrower, or because of information problems. It is the latter on which we shall concentrate here.

As has been argued above, banks must assess the credit-worthiness of borrowers without having a foolproof basis for risk assessment; the probability of default and the future value of collateral are both subject to uncertainty. It has long been understood that small firms pose particular problems in this regard because of their relative absence of track-record and because the scale of borrowing may not justify the costs of information gathering.

A succession of reports on the UK has suggested that there is a credit gap for small firms (Macmillan, 1931; Radcliffe, 1959; Bolton, 1971; Wilson, 1979). As a result, various programmes were introduced to fill the gap. The most recent study (Department of Trade and Industry (DTI), 1991) concludes that there is no longer a significant gap. The first part of the study consists of the results of a survey of small firms, and identifies very little evidence of market failure either on the supply side (in terms of information acquired) or the demand side (in terms of information supplied). However, a somewhat different picture emerges from between the lines of the second part of the study, which is based on a survey of financial institutions. The scope of this survey is greater than the first because it reveals information on treatment of all loan applicants, not just those which have survived. While 95% of respondents cited inadequate security, and firms already at their borrowing limit, as frequent reasons for rejecting applications, 75% cited poor presentation, lack of professional input and lack of track record. The conclusion is drawn 'that management ability is the most significant and pervasive constraint on small firms' (DTI, 1991, p. 123). But the possibility that problems may also arise from the lenders' side is evident from the subsequent passage:

There are undoubtedly also some instances of funding problems arising from a range of market failures. Those failures related to poor or inaccurate information and to incorrect perceptions which should be addressable by well-targeted public sector action. Problems stemming from institutional attitudes and practices are less readily addressed. (DTI, 1991, p. 123)

This passage supports the argument being developed in this paper, that the concept of market failure is only of limited relevance when analysing market behaviour under uncertainty. It is in the nature of small firms, particularly the more innovative ones, that the maximum amount of relevant information which could in principle be collected is still too low to attach great weight to any prediction of credit-worthiness. Not only do they lack a track record, but they also generally operate in highly competitive markets where there is little scope for defensive strategies. This contrasts with larger, well-established firms which generally enjoy a degree of market power and an adequate asset base to protect them in conditions which might threaten their capacity to service loans. As Minsky (1982, ch. 5) elaborates, debt servicing is not just a matter of actual and expected returns, but also of cash-flow; small firms with low asset bases are particularly vulnerable to interest rate hikes.

The importance of cash flow has been picked up in the empirical US literature on credit rationing. Fazzari, Hubbard and Patersen (1988) tested the hypothesis that there are significant capital market imperfections for firms which are not large and mature. The test (covering the period 1970–1984) involved correlating investment with cash flow, correcting for the indications of expected returns on investment contained in cash-flow information. They show that, for firms which habitually finance investment with retained earnings, investment does appear to be constrained by cash flow; the presumption is that these firms perceive a higher cost from outside finance than retained earnings (i.e. there is capital market imperfection). Further studies along these lines are gathered in

Hubbard (1990). The evidence incidentally supports the argument that credit rationing is also a cyclical phenomenon.

There remains the question of how far non-availability of credit accounts for the choice to finance internally, rather than relative cost of internal and external finance. Morgan (1991) distinguishes in his US study (1980–1984 data) between those firms which have a credit commitment (overdraft facility) and those which do not. He finds that investment was constrained twice as much on average by reductions in cash flow for firms without commitments than for those with commitments, although their investment prospects (measured by q) were more than twice as high. Morgan points out that banks have been found to be willing to give commitments to firms according to the extent of their experience of them as borrowers and the number of years they have been in business. Yet for small firms credit commitments are particularly valuable, not only as a direct source of credit, but as a signal to other lenders of their credit-worthiness. A 1990 Federal Reserve survey quoted by Morgan found that 73% of small firms did not have a commitment, compared with 40% of medium-sized firms. Moore (1988, p. 25) points out that over 70% of new loans to industrial and commercial companies in the US are made under commitments. While there is no explicit evidence that relatively lower use of commitments by small and medium-sized firms is due to supply constraints, it is nevertheless reasonable to presume that such is the case. In other words, significant rationing of small firms may well occur at the stage of application to banks for commitments.

Fazzari *et al.* (1988) conclude that it is important to understand the accumulation of 'information capital' through financial intermediation, because of the consequence for investment of information imperfections in capital markets. Again, the implicit benchmark is full information. But full information for risk assessment of any firm is not even available in principle, far less in practice. For small firms, the lack of information is of necessity more severe than for larger, mature firms. There are well-established conventions for risk assessment of the latter type of firm, which allow risk to be measured without adequate information, as must always be the case, given the uncertainty about the future. The problems of small firms are compounded by the fact that the conventions are lacking. For them it is not in general the case that the banks are price setters and quantity takers.

Conclusion

It has been argued here that there is systematic bank rationing of credit. This is evident in the cyclical pattern of valuation of collateral for potential bank borrowers and in the somewhat lagged cyclical valuation of banks' own credit-worthiness. Both serve to constrain credit availability during cyclical downturns. It has also been argued that particular classes of borrowers experience systematic credit rationing: notably small businesses.

The reasons for this rationing have been distinguished from the leading explanation for 'equilibrium rationing': information asymmetries. Implicit in this latter explanation is the presumption that full information and a full risk assessment are possible. But Keynes's epistemology demonstrates that in general there can be no full basis for risk assessment. Greater weight might be attached to some assessments than to others, because of a greater amount of information. But an estimate of risk (other than in circumstances where structure can be treated as stable, as in actuarial risk assessment), must rely

ultimately on convention. Different conventions have evolved for the risk-assessment of small firms which must account at least partially for their experience of credit rationing; their rationing cannot be ascribed to differences in objective risk, since such cannot be measured.

But, more generally, banks vary their willingness to extend credit on the basis of their expectation of the viability of investment projects and the value of collateral. These expectations are formed from a combination of information, animal spirits and convention, all of which can account for a counter-cyclical pattern of rationing. The underlying process is still one of a credit-driven monetary system, with limited influence from the central bank. However, by incorporating an active role for markets in determining bank capital, and for banks in determining risk-assessment and their own liquidity preference, endogenous credit theory is enriched. The theory can then demonstrate an active role of finance in the business cycle, in industrial structure (and thus growth potential), and in international income distribution.

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